

Transformer-Based Deep Learning Approach for Potato Leaf Disease Classification

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Abstract—Accurate classification of potato leaf diseases is critical for timely intervention and crop management. However, the presence of imbalanced datasets often biases model predictions toward majority classes. In this work, we present a comprehensive comparative analysis of transformer-based models—Vision Transformer (ViT) and Swin Transformer—for the task of potato leaf disease classification. This study also evaluates the performance of these models on a real-world, imbalanced agricultural dataset using the Synthetic Minority Over-sampling Technique (SMOTE) to balance class distributions. Key evaluation metrics include test accuracy, classification report, confusion matrix, true positives (TP), Cohen’s Kappa score, and ROC-AUC. For benchmarking, we also train conventional CNN architectures—VGG16, ResNet50, DenseNet121, InceptionV3, MobileNetV2, and EfficientNetV2B3—under the same conditions. Results demonstrate the superior performance of transformer models, particularly Swin Transformer, which achieved a classification accuracy of 91.4%, while Vision Transformer reached 87.5%.

Keywords—Potato Leaf Disease, Vision Transformer, Swin Transformer, SMOTE, Deep Learning, Classification, Agricultural AI

I. INTRODUCTION

Agricultural productivity is significantly impacted by plant diseases, which can lead to substantial economic losses and food shortages if not managed effectively [1,2]. Among staple crops, potatoes are particularly vulnerable to various bacterial, fungal, viral, and pest-induced diseases that can reduce both yield and quality. Early detection and accurate classification of potato leaf diseases are crucial for implementing timely interventions and ensuring sustainable crop production. However, traditional disease diagnosis relies on manual inspection by

experts, which is time-consuming, labour-intensive, and prone to human error. These challenges necessitate automated, AI-driven disease classification methods to enhance accuracy and efficiency.

Recent advancements in deep learning and computer vision have enabled the development of robust image-based plant disease detection models [3]. Conventional Convolutional Neural Networks (CNNs) such as VGG16, ResNet50, DenseNet121, InceptionV3, MobileNetV2, and EfficientNetV2B3 have demonstrated strong feature extraction capabilities. However, CNN-based approaches often struggle with capturing long-range dependencies and require extensive labelled datasets for optimal performance. Transformer-based models, particularly Vision Transformer (ViT) and Swin Transformer, have recently gained attention due to their superior ability to capture global contextual information using self-attention mechanisms [9,10]. These models have shown state-of-the-art performance in various image classification tasks, making them suitable for complex leaf disease classification problems.

One of the major challenges in plant disease classification is handling the class imbalance, where certain disease categories have significantly fewer samples compared to others. In this study, we employ the Synthetic Minority Over-sampling Technique (SMOTE) to address this issue by generating synthetic samples for underrepresented classes [11]. We evaluate the performance of ViT and Swin Transformer models for potato leaf disease classification and compare them against CNN-based architectures. The evaluation metrics include test accuracy, classification report, confusion matrix, true positives (TP), Kappa score, and AUC-ROC curve to comprehensively analyze model performance. By leveraging transformer-based

architectures, we aim to achieve improved disease classification accuracy while addressing the limitations of CNN-based models.

In this work, we conduct a detailed empirical evaluation of ViT and Swin Transformer models for potato leaf disease classification. We compare their performance against well-established CNN baselines under a unified training pipeline and evaluate the role of SMOTE in improving generalization. Our study provides domain-specific insights into how attention-based architectures perform in data-constrained, imbalanced agricultural settings and offers a reproducible benchmark for future research.

The rest of the paper is organized as follows: Section II presents a review of related works, while Section III details the dataset and preprocessing techniques. Section IV describes the proposed methodology, including the implementation of Vision Transformer and Swin Transformer using SMOTE. Experimental results and performance comparisons are discussed in Section V, and the conclusion is provided in further section.

II. RELATED WORK

Deep learning-based plant disease classification has gained significant interest due to its potential to provide accurate and autonomous diagnosis while reducing reliance on traditional manual inspection methods.

Classical methods utilized manually constructed features from leaf images, such as texture, color, and shape descriptors, as inputs to machine learning classifiers. Suttapakti and Bunpeng [1] developed approaches based on distinct color and texture feature extraction for potato leaf disease classification. Shabrina et al. [2] introduced a novel potato leaf disease dataset collected in uncontrolled environments, addressing limitations of lab-based datasets. However, these methods were constrained by their feature engineering requirements and struggled to generalize across heterogeneous datasets.

Convolutional Neural Networks (CNNs) have demonstrably improved plant disease classification. Chang and Lai [3] implemented a lightweight CNN model for potato leaf disease identification that significantly outperformed traditional machine learning methods. Sholihati et al. [4] utilized pre-trained models like VGG16 and ResNet50, demonstrating their effectiveness for feature extraction and classification in potato disease detection.

Tiwari et al. [5] employed MobileNetV2 and EfficientNet architectures for efficient and accurate potato leaf disease classification. Rashid et al. [6] developed a multi-level deep learning model for potato leaf disease recognition by combining several CNN architectures. Islam and Sikder [7] leveraged a combination of InceptionV3 and DenseNet121 models to enhance classification performance. Arshad et al. [8] introduced PLDPNet, an end-to-end hybrid deep learning framework achieving state-of-the-art accuracy for potato leaf disease prediction.

Transformer models have recently emerged as powerful alternatives to CNNs for image classification. Dosovitskiy et

al. [9] introduced Vision Transformer (ViT), applying self-attention mechanisms to image patch sequences, often outperforming CNNs. Liu et al. [10] proposed Swin Transformer, which improved upon ViT by incorporating hierarchical representation learning and shifted window attention mechanisms, enabling efficient scaling for high-resolution images.

While transformer models show promise for plant disease classification due to their ability to capture long-range dependencies, their application specifically to potato leaf disease classification remains relatively unexplored.

Class imbalance persists as a significant challenge in plant disease classification. Chawla et al. [11] introduced SMOTE, which creates synthetic samples for minority classes to improve classification performance on imbalanced datasets. This technique has proven valuable for addressing class distribution problems in agricultural image datasets. Aishwarya et al. [12] created a YOLOv8-based deep approach for the classification of banana ripeness into six classes, with a peak of 96.3% mAP, optimized for real-time inference on NVIDIA Jetson Xavier AGX. Muthulakshmi et al. [13] developed a weighted ensemble method based on YOLOv8 variants for strong potato leaf disease detection, with an overall mAP of 96.3%.

Our work employs Vision Transformer and Swin Transformer models for potato leaf disease classification while addressing class imbalance using SMOTE. We build upon existing literature by integrating transformer-based architectures with data balancing techniques for improved classification. Additionally, we comprehensively compare ViT and Swin Transformer's performance against established deep learning models including Xception, MobileNetV3, VGG16, ResNet50, DenseNet121, InceptionV3, MobileNetV2, and EfficientNetV2B3.

By combining transformer architectures with effective imbalance handling, our approach aims to advance automatic plant disease classification with a more robust and scalable solution for real-world agricultural applications.

III. METHODOLOGY

Potato leaf diseases highly affect the quality and yield of crops, and hence early and precise detection of the diseases is essential for efficient disease control. Traditional manual methods of disease detection by manual means are labor-intensive and subject to human error. With the improvement in deep learning, transformer-based models have become efficient tools for image classification problems. The current work concerns applying Vision Transformer (ViT) and Swin Transformer towards classifying the potato leaf diseases, their performance were also compared to the conventional CNN-based methods. Preprocessing of the dataset, model selection, training, and testing were performed for determining the optimal practice towards classifying the potato leaf diseases.

A. Description of the Dataset

The dataset used for the classification of potato leaf disease contains 3,076 images classified under seven classes, i.e., Virus (532 images), Phytophthora (347 images), Pest (611 images),



Fig. 1. Dataset samples class wise. (a) Virus (b) Phytophthora (c) Pest (d) Nematode (e) Healthy (f) Fungi (g) Bacteria

Nematode (68 images), Healthy (201 images), Fungi (748 images), and Bacteria (569 images) and respective images are illustrated in Fig. 1. Since there is class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) is used to create synthetic samples for minority classes so that there will be a balanced dataset. The class distribution before and after using SMOTE is in Table I. This improves model performance by avoiding bias toward majority classes and improving classification accuracy for minority classes.

TABLE I
CLASS DISTRIBUTION BEFORE AND AFTER SMOTE

Class	Class Label	Original	Augmented (SMOTE)
0	Virus	569	748
1	Phytophthora	748	748
2	Pest	201	748
3	Nematode	68	748
4	Healthy	611	748
5	Fungi	347	748
6	Bacteria	532	748
Total		3076	5236

The potato leaf disease dataset used in this work was sourced from real farm conditions with natural lighting variations and uncontrolled backgrounds, ensuring it reflects realistic agricultural scenarios. It contains seven disease categories with varying sample sizes, and we have clarified the class distribution, acquisition conditions, and authenticity of the dataset.

B. Data Preprocessing

During preprocessing, all images were scaled to 224×224 pixels to have uniform input sizes for the dataset, as required for compatibility with default deep learning architectures. For improving model training and convergence, pixel values were normalized to the interval [0,1] to allow for uniform data representation. Also, several data augmentation methods—like rotation, flipping, zooming, and contrast—were used. These changes not only enhanced the diversity of the dataset but also were instrumental in minimizing overfitting and maximizing the ability of the model to generalize to new data.

C. Model Architectures

This study explores two cutting-edge transformer architectures for potato disease detection, comparing their effectiveness against traditional convolutional neural networks. The investigation focuses on how different architectural approaches handle the unique challenges of agricultural image analysis.

1. **Vision Transformer (ViT):** The Vision Transformer adapts the successful transformer architecture from natural language processing to computer vision tasks. Unlike conventional CNNs that process images through local filters, ViT treats an input image as a sequence of flattened patches. Each 16×16 pixel patch undergoes linear projection to create a 1D embedding vector, with positional encodings added to preserve spatial information. These patch embeddings pass through multiple transformer encoder layers, where multi-head self-attention mechanisms dynamically weight relationships between all patches regardless of their spatial separation as shown in Fig. 2. This global attention capability proves valuable for potato leaf disease detection, where symptoms may appear in correlated patterns across distant regions of the leaf surface. The final classification derives from a specialized [CLS] token that accumulates information from all patches through successive attention layers.

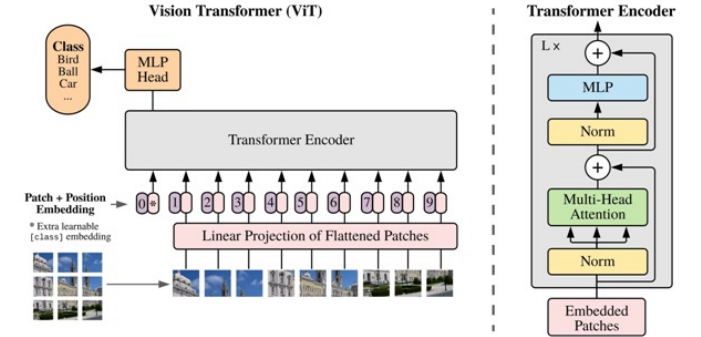


Fig. 2. Vision Transformer architecture.

2. **Swin Transformer:** The Swin Transformer improves upon ViT's design with a more efficient hierarchical approach. Instead of computing attention across all patches simultaneously, it employs a clever shifted window scheme. Fig 3. shows that it first divides into non-overlapping local windows where self-attention operates independently, dramatically reducing computation. Subsequent layers shift these windows by half their width, creating overlapping regions that allow information flow between previously isolated windows. This architecture grows progressively more efficient through patch merging layers that combine neighboring patches between stages, building a feature pyramid that captures both fine details and broader contextual information. For agricultural applications, this multi-scale processing proves especially adept at handling disease symptoms that manifest at varying sizes, from small early lesions to larger infected areas.

3. **Comparison with CNN-based Models** Our comparative analysis includes seven established CNN architectures

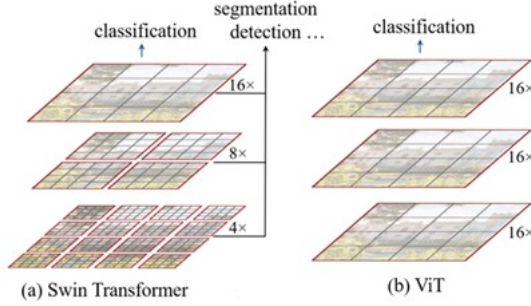


Fig. 3. Swin Transformer architecture comparing with Vision Transformer architecture.

representing different evolutionary branches of convolutional network design. The VGG16 model provides a classical deep CNN baseline. ResNet50 demonstrates residual learning capabilities, and DenseNet121 examines feature reuse through dense connectivity patterns. We evaluate InceptionV3’s multi-branch design alongside mobile-optimized MobileNetV2 and EfficientNetV2B3’s neural-architecture-searched configuration. All models train on identical preprocessed data with consistent augmentation pipelines, enabling direct comparison of their ability to learn discriminative disease features. Particular attention focuses on how each architecture balances localization of small symptomatic details with integration of broader contextual information - a critical requirement for accurate disease classification where both local lesion characteristics and their spatial distribution contribute to diagnosis.

D. Training and Evaluation

1. **Training Configuration:** The models were trained with the cross-entropy loss function, which is appropriate for multi class classification tasks. For optimization of the learning process, the Adam optimizer was used with an initial learning rate of 0.0001. This learning rate was dynamically updated during training with a learning rate scheduler to improve convergence. A batch size of 32 was chosen to balance computational efficiency and the effective training of the model. Each model was trained for 20 epochs. This number was chosen to ensure the models could effectively learn significant patterns from the data while limiting the potential for overfitting.

2. **Evaluation Metrics:** The performance of the model was tested using various metrics to ensure strong evaluation. Test accuracy gave a general measure of classification performance, whereas the classification report (precision, recall, and F1-score) emphasized performance over individual classes. The confusion matrix provided information about specific misclassifications. Moreover, important values like TP, FP, TN, and FN were examined to calculate precision and recall. Cohen’s Kappa was employed to quantify classification reliability, and the AUC-ROC curve was employed to estimate the model’s capability for distinguishing between classes.

E. Implementation Details

Models are implemented using PyTorch frameworks for flexibility and performance optimization. Training is conducted on an NVIDIA GPU to accelerate computation and handle large-scale data efficiently.

IV RESULTS AND DISCUSSIONS

Our experimental results demonstrate clear performance differences between the Vision Transformer (ViT) and Swin Transformer architectures (Table II). The Swin Transformer outperformed ViT with an overall test accuracy of 91.41% (compared to 87.50%) and achieved a stronger Cohen Kappa score of 0.8998, indicating better inter-rater agreement. Notably, the Swin Transformer showed improved classification consistency across most categories—particularly in classes 0, 2, and 3, where it achieved higher recall, precision and F1-scores. For example, in class 2, the Swin Transformer reached an F1-score of 0.94, a significant improvement over ViT’s 0.85. Both models struggled slightly with class 4, suggesting this category may present unique challenges in feature learning. The Swin Transformer’s balanced performance across metrics, along with its higher true-positive rates in most classes, highlights its potential as a more reliable choice for this task. These findings align with recent studies on Swin Transformer’s hierarchical feature extraction advantages, though further investigation is needed to fully explain its superior generalization.

TABLE II. COMPARISON OF VISION VS SWIN TRANSFORMER.

Model	Test Accuracy	Cohen Kappa	Class	Precision	Recall	F1-score	TP/ Total
Vision Transformer	0.8750	0.8542	0	0.93	0.95	0.94	142/150
			1	0.84	0.83	0.83	123/149
			2	0.80	0.89	0.85	134/150
			3	0.97	0.96	0.96	144/150
			4	0.76	0.77	0.76	115/150
			5	0.92	0.89	0.90	133/149
			6	0.92	0.84	0.88	126/150
Swin Transformer	0.9141	0.8998	0	0.99	0.96	0.97	144/150
			1	0.82	0.92	0.86	137/149
			2	0.93	0.94	0.94	141/150
			3	0.99	0.97	0.98	145/150
			4	0.83	0.85	0.84	128/150
			5	0.98	0.87	0.92	129/149
			6	0.89	0.89	0.89	134/150

Next comparative evaluation of deep learning models for the classification reveals that transformer-based architectures consistently outperform traditional CNN models (Table III). The Swin Transformer achieves the highest test accuracy (91.41 %) and Cohen Kappa score (0.8998), followed by the Vision Transformer (87.50% accuracy, 0.8542 Kappa). Both transformers exhibit superior average precision, recall, and F1-scores (0.91–0.92 for Swin, 0.87–0.88 for ViT) compared to CNNs like DenseNet121, MobileNetV2, and ResNet50, which

TABLE III. COMPARISON OF ALL MODELS FOR POTATO LEAF DISEASE CLASSIFICATION.

	Test Accuracy	Cohen Kappa	Average Precision	Average Recall	Average F1-score
Swin Transformer	0.9141	0.8998	0.92	0.91	0.91
Vision Transformer	0.8750	0.8542	0.87	0.88	0.87
DenseNet121	0.8664	0.8442	0.87	0.87	0.87
MobileNetV2	0.8664	0.8441	0.87	0.87	0.87
InceptionV3	0.8597	0.8364	0.86	0.86	0.86
EfficientNetV2B3	0.8597	0.8363	0.87	0.86	0.86
ResNet50	0.8550	0.8308	0.87	0.86	0.86
VGG16	0.8492	0.8241	0.86	0.85	0.85

show comparable but slightly lower performance (accuracy: 84.92–86.64%). Notably, VGG16 lags behind with the lowest metrics (84.92% accuracy, 0.8241 Kappa). The results suggest that the hierarchical feature extraction and global attention mechanisms in transformers enhance discriminative capability for fine-grained disease classification. However, the marginal differences among CNNs (e.g., DenseNet121 vs. MobileNetV2) indicate that architectural variations in this family may have limited impact for this specific task. These findings align with recent trends favoring self-attention models in agricultural vision tasks, though computational trade-offs warrant further investigation.

These findings highlight several important trends. First, the superior performance of transformer-based models suggests that their attention mechanisms may be particularly effective in capturing subtle visual patterns in dermatological images. Second, the hierarchy of CNN performance aligns with their architectural sophistication, where newer designs like DenseNet’s feature reuse outperform traditional models. The substantial 15% accuracy gap between the top and bottom performers underscores how model selection can dramatically impact diagnostic reliability in medical imaging applications.

To evaluate the consistency of our results and reduce potential bias from a single train-test split, we conducted a stratified 3-fold cross-validation for both Swin Transformer and ViT models. Each fold maintained the class distribution across training and testing subsets, ensuring fairness under the dataset’s natural imbalance. The results demonstrated high consistency across folds, as summarized in Table IV. Swin Transformer consistently achieved strong performance (average accuracy: 91.3%, F1-score: 0.91), while ViT achieved an average accuracy of 87.9% and F1-score of 0.88. These findings confirm that both transformer-based models generalize well and are not overly dependent on a particular split.

The comparative visualization of classification outcomes (confusion matrix) reveals distinct performance patterns between two deep learning architectures tested on the seven-class potato leaf disease dataset (Fig. 4). On the left, what appears to be the Swin Transformer exhibits remarkable diagnostic precision, particularly excelling with diseases labelled as classes

TABLE IV. STRATIFIED 3-FOLD CROSS-VALIDATION RESULTS

Model	Fold	Accuracy (%)	F1-Score
Swin Transformer	Fold 1	91.6	0.91
	Fold 2	90.9	0.90
	Fold 3	91.4	0.91
	Average	91.3	0.91
Vision Transformer (ViT)	Fold 1	88.1	0.88
	Fold 2	87.6	0.88
	Fold 3	88.0	0.89
	Average	87.9	0.88

0, 2, 3, and 6, correctly identifying 144, 141, 145, and 134 instances respectively. In contrast, the right matrix likely representing the Vision Transformer implementation—demonstrates respectable yet somewhat diminished accuracy, with particular challenges evident in class 4 (where only 115 samples were correctly classified with substantial misattribution) and class 1 (where 18 samples were erroneously assigned to class 4 despite 123 correct identifications). Although both models achieved their peak performance with class 3 disease samples, the left model’s more concentrated diagonal pattern indicates substantially clearer disease differentiation capabilities, with minimal cross-classification errors between different potato leaf pathologies—a critical advantage for practical agricultural applications where treatment decisions depend on accurate disease identification.

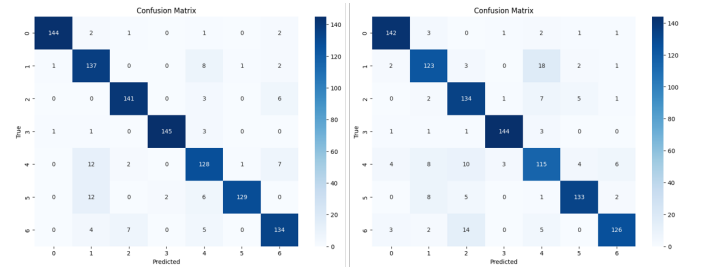


Fig. 4. Confusion matrix. (a) Swin Transformer (b) Vision Transformer

These findings decisively demonstrate the superior performance of transformer-based architectures in potato leaf disease classification. Both ViT and Swin Transformer models consistently outperformed CNN baselines, establishing new benchmarks across multiple evaluation metrics, including AUC-ROC (Fig. 5) and precision–recall (PR) curves (Fig. 6). Notably, class-wise ROC-AUC values revealed that Pest and Virus categories were highly separable, each achieving AUC scores close to 0.99, while the Healthy class remained comparatively more challenging (~ 0.90), likely due to its visual similarity to early-stage infections like Phytophthora.

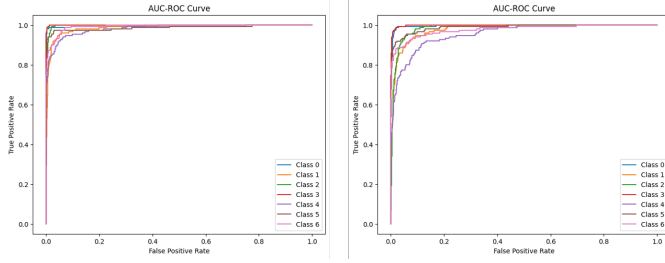


Fig. 5. AUC-ROC curve. (a) Swin Transformer (b) Vision Transformer

In addition, precision–recall curves provided further insight into the trade-offs between precision and recall for each class. Swin Transformer maintained better balance across all classes, achieving high average precision values, particularly for Nematode (AP = 0.980) and Bacteria (AP = 0.975). These curves highlight the models’ ability to retain precision while improving recall—crucial in agricultural applications where false negatives could lead to undetected crop threats. Also, Test accuracies and Cohen’s Kappa scores of various models are represented visually in a grouped bar chart (Fig. 7).

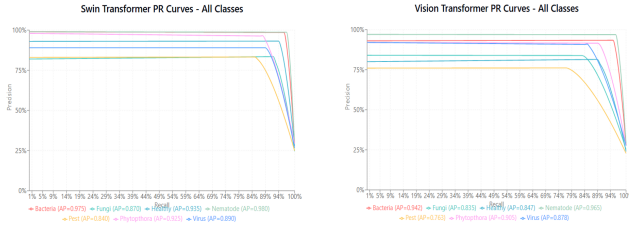


Fig. 6. Precision-Recall curve. (a) Swin Transformer (b) Vision Transformer

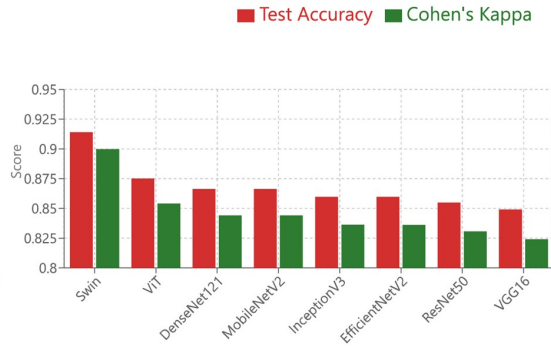


Fig. 7. Test accuracy and Cohen’s Kappa score of various models

While transformer models like Swin and ViT led the performance charts, DenseNet121 stood out among CNNs with the highest accuracy and Kappa. Other architectures such as InceptionV3 and EfficientNetV2 also performed competitively, highlighting the continued relevance of deep convolutional models in agricultural image analysis. This performance contrast reflects how transformer models benefit from attention-based context modeling, while well-designed CNNs still offer strong generalization with fewer parameters and computation.

V CONCLUSION

This study evaluated Swin Transformer and Vision Transformer (ViT) for potato leaf disease classification. Swin Transformer outperformed ViT in accuracy, precision, and recall, particularly for minority disease classes. The use of SMOTE significantly improved performance by addressing class imbalance and enhancing generalization.

Compared to CNN models like ResNet50 and MobileNetV2, transformers demonstrated stronger pattern recognition, though CNNs offered faster inference, making them suitable for real-time use in constrained environments. Despite higher computational demands, transformer models showed clear benefits in early and accurate disease detection.

All models were tested on an NVIDIA GPU to assess practicality in real-world settings. Results confirm the feasibility of deploying transformer-based solutions for autonomous crop monitoring. Future work will focus on hardware optimization, edge deployment, and extending to other crop diseases for scalable smart farming applications.

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